

Evaluating Limits

ANSWER KEY



Direct substitution and algebraic techniques

1 Evaluate each limit:

a $\lim_{x \rightarrow 1} (3x^2 - 2x + 5) = 6.$

b $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = 4.$

c $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$ Rationalize: $\frac{(x+4) - 4}{x(\sqrt{x+4} + 2)} = \frac{1}{\sqrt{x+4} + 2} \rightarrow \frac{1}{4}.$

2 Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x^2 - 9}.$

Factor: $\frac{(x-3)(x+2)}{(x-3)(x+3)} = \frac{x+2}{x+3} \rightarrow \frac{5}{6}.$

Trigonometric and special limits

3 Using $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, evaluate:

a $\lim_{x \rightarrow 0} \frac{\sin 5x}{x} = 5 \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} = 5.$

b $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$ (known result: this limit is 0, not ∞).

c $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x} = \frac{3}{5} \cdot \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot \frac{5x}{\sin 5x} = \frac{3}{5}.$

4 Evaluate $\lim_{x \rightarrow \infty} \frac{5x^3 - 2x + 1}{2x^3 + 7}.$

Equal degrees: the limit is the ratio of leading coefficients, $\frac{5}{2}.$

Limits that fail to exist and the squeeze theorem

5 Explain why $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ does not exist.

As $x \rightarrow 0$, $\frac{1}{x} \rightarrow \pm \infty$, so $\sin(1/x)$ oscillates between -1 and 1 infinitely often near $x = 0$ rather than approaching a single value.

6 Use the squeeze theorem to evaluate $\lim_{x \rightarrow 0} x^2 \cos\left(\frac{1}{x}\right).$

Since $-x^2 \leq x^2 \cos(1/x) \leq x^2$ and $\lim_{x \rightarrow 0} (\pm x^2) = 0$, the squeeze theorem gives a limit of 0.

Limits with absolute value and piecewise functions

7 Evaluate $\lim_{x \rightarrow 3} |x - 3| \cdot \frac{1}{x}.$

As $x \rightarrow 3$, $|x - 3| \rightarrow 0$ and $\frac{1}{x} \rightarrow \frac{1}{3}$ (a finite value), so the product limit is $0 \cdot \frac{1}{3} = 0.$

- 8 For $f(x) = 2x + 1$ if $x < 3$, and $f(x) = x^2 - 2$ if $x \geq 3$, evaluate $\lim_{x \rightarrow 3^-} f(x)$ and $\lim_{x \rightarrow 3^+} f(x)$, and state whether $\lim_{x \rightarrow 3} f(x)$ exists.
 $\lim_{x \rightarrow 3^-} f(x) = 2(3) + 1 = 7$. $\lim_{x \rightarrow 3^+} f(x) = 3^2 - 2 = 7$. Since both one-sided limits equal 7, $\lim_{x \rightarrow 3} f(x) = 7$ exists.

More algebraic limit techniques

- 9 Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$ by factoring with the difference-of-cubes formula.
 $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$, so the limit is $\lim_{x \rightarrow 2} (x^2 + 2x + 4) = 4 + 4 + 4 = 12$.
- 10 Evaluate $\lim_{h \rightarrow 0} \frac{(2+h)^2 - 4}{h}$.
 $(2+h)^2 - 4 = 4 + 4h + h^2 - 4 = 4h + h^2 = h(4 + h)$, so the limit is $\lim_{h \rightarrow 0} (4 + h) = 4$.

Visualizing a limit numerically and graphically

- 11 The function $f(x) = \frac{\sin x}{x}$ (undefined at $x = 0$) is graphed below. Use the graph to estimate $\lim_{x \rightarrow 0} f(x)$.
The graph approaches height 1 from both sides as $x \rightarrow 0$, even though the function has a hole exactly at $x = 0$: $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

Limits at infinity with different degree relationships

- 12 Evaluate each limit at infinity, based on the relative degrees of numerator and denominator:
- a $\lim_{x \rightarrow \infty} \frac{4x+3}{x^2-1}$ (lower degree numerator) $= 0$ (denominator grows faster).
- b $\lim_{x \rightarrow \infty} \frac{x^3-1}{2x^2+5}$ (higher degree numerator) $= \infty$ (numerator grows faster; the limit does not exist as a finite number).

Limits requiring conjugate multiplication

- 13 Evaluate $\lim_{x \rightarrow 9} \frac{x-9}{\sqrt{x}-3}$ by rationalizing the denominator.
Multiply by $\frac{\sqrt{x}+3}{\sqrt{x}+3} \cdot \frac{(x-9)(\sqrt{x}+3)}{x-9} = \sqrt{x} + 3 \rightarrow \sqrt{9} + 3 = 6$.
- 14 Evaluate $\lim_{x \rightarrow -1} (\sqrt{x^2 + 3x} - x)$ by rationalizing.
Multiply by $\frac{\sqrt{x^2 + 3x} + x}{\sqrt{x^2 + 3x} + x}$: $\frac{3x}{\sqrt{x^2 + 3x} + x}$. Dividing top and bottom by x : $\frac{3}{\sqrt{1 + 3/x} + 1} \rightarrow \frac{3}{1+1} = \frac{3}{2}$.

Piecewise limits requiring a parameter

- 15 For $f(x) = kx + 1$ if $x \leq 2$, and $f(x) = x^2 - 3$ if $x > 2$, find k so that $\lim_{x \rightarrow 2} f(x)$ exists.
Need $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$: $2k + 1 = 4 - 3 = 1$, so $k = 0$.

Limits involving factorable higher-degree polynomials

- 16** Evaluate $\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1}$ by factoring the numerator as a difference of squares twice.
 $x^4 - 1 = (x^2 - 1)(x^2 + 1) = (x - 1)(x + 1)(x^2 + 1)$, so the limit is $\lim_{x \rightarrow 1} (x + 1)(x^2 + 1) = (2)(2) = 4$.
- 17** Evaluate $\lim_{x \rightarrow -2} \frac{x^3 + 8}{x + 2}$ using the sum-of-cubes factoring pattern.
 $x^3 + 8 = (x + 2)(x^2 - 2x + 4)$, so the limit is $\lim_{x \rightarrow -2} (x^2 - 2x + 4) = 4 + 4 + 4 = 12$.
- 18** Evaluate $\lim_{x \rightarrow 0} \frac{(3+x)^2 - 9}{x}$.
 $(3 + x)^2 - 9 = 9 + 6x + x^2 - 9 = 6x + x^2 = x(6 + x)$, so the limit is $\lim_{x \rightarrow 0} (6 + x) = 6$.